

Static-to-Adaptive Phase Structure of Five-Qubit Logical Bell Measurements under Photon Erasure

Exact $q = 2/3$ Basis Switching, Feedforward Optimality, and Failure-Record Gauge Equivalence

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Abstract

We determine the complete static-to-adaptive photon-erasure phase structure of logical Bell measurements for the non-CSS five-qubit $[[5, 1, 3]]$ perfect code within transversal ancilla-free $X/Y/Z$ -guaranteed linear-optical Bell analyzers. In the static grammar, each physical pair has six analyzer choices. Exact exhaustion of all $6^5 = 7776$ schemes collapses the search space to four rational loss-response polynomials. Only two reach the upper envelope, and their difference factors as

$$P_H(q) - P_L(q) = \frac{7}{32}q^4(3q - 2),$$

forcing an exact basis switch at intact-pair survival $q_c = 2/3$, equivalently single-photon survival $\eta_c = \sqrt{2/3}$. The high-survival class consists precisely of the 18 full-support representatives of nontrivial logical Pauli operators in the five-qubit normalizer, times the 32 choices of resolved Bell halves. We then solve the adaptive problem exactly. Two independent dynamic programs generate the same 49 312 information states and the same optimum

$$P_{\text{ad}}^*(q) = \frac{5}{4}q^3 + \frac{7}{4}q^4 - \frac{65}{32}q^5,$$

with survivor-sector probabilities $(0, 0, 0, 1/8, 17/20, 31/32)$. One feedforward policy is optimal for every $q \in [0, 1]$: all 313 200 Bellman action comparisons reduce to twelve elementary nonnegative gap polynomials. Consequently adaptivity does not move the static $q = 2/3$ boundary; it removes it, strictly dominating the static envelope for $0 < q < 1$. Finally, a doubled-stabilizer gauge lemma proves that the sign-resolved local failure record of a physical Bell analyzer cannot improve the adaptive optimum. Independent affine, symplectic, and full-instrument replays support every released identity. The result is a finite-code, proof-carrying phase theorem inside a declared optical grammar, not a universal linear-optical loss-threshold theorem.

1 Introduction

Bell-state measurements (BSMs) are central to quantum teleportation, entanglement swapping, quantum repeaters,

measurement-based quantum computation, and fusion-based photonic architectures [3, 4, 5, 6, 7]. For two dual-rail photonic qubits, passive ancilla-free linear optics cannot unambiguously distinguish all four Bell states with probability exceeding $1/2$ [7]. Encoding and additional optical resources can raise the logical success probability under different resource assumptions [8, 9, 18, 19].

Logical BSMs become especially important when quantum error correction is already part of the architecture. Static photon-loss optimization has been developed for CSS codes [20], general static/adaptive loss-tolerance bounds are known [24], and arbitrary stabilizer codes, including the five-qubit code, have recently been treated in the ideal lossless setting [27]. A two-qubit repetition-code logical BSM has also been demonstrated experimentally beyond the bare 50% linear-optical ceiling by Kumar, Reiss, van Loock, and Barz [28].

Relation to the recent experiment. Kumar, Reiss, van Loock, and Barz recently demonstrated the practical feasibility of encoded photonic Bell analysis using a two-qubit repetition/quantum-parity encoding [28]. The present work addresses a different question that becomes natural once one moves from that repetition encoding to a genuine distance-three non-CSS error-correcting code: how should the physical Bell-measurement architecture itself change under photon erasure, and can adaptive measurement choice remove a finite-resource measurement transition? For the five-qubit $[[5, 1, 3]]$ code we answer those questions exactly. We obtain an exhaustive static four-class collapse and an algebraic basis switch at $q = 2/3$, then prove that adaptive feedforward removes that switch entirely while preserving the lossless $31/32$ endpoint. Thus the experiment supplies an important motivating proof of feasibility, whereas the results established here are finite-code optimization and structure theorems for a different code, loss model, and control problem.

The five-qubit perfect code is the smallest code correcting an arbitrary single-qubit error [13]. Because it is non-CSS, the finite-erasure optimization of all transversal $X/Y/Z$ -guaranteed physical Bell analyzers is not supplied directly by the CSS analysis. The present paper asks two linked questions. First, which *static* physical-BM architecture maximizes logical success as photon survival varies? Second, if the next measurement may depend on earlier erasures and outcomes, does the static transition survive?

The answer is a complete static-to-adaptive phase story. Static optimization has an exact basis switch at $q =$

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2/3. The switch is caused by a tradeoff between one-erasure robustness and fully intact performance. Adaptive feedforward removes the tradeoff: it preserves the optimal intact sector while raising the one-erasure sector to 17/20. The resulting adaptive law lies strictly above the static envelope throughout the lossy interior.

Claim boundary. We do not claim a new five-qubit code, a new bare optical Bell limit, a new universal photon-loss threshold, or novelty for adaptive logical BSMs as such. The contribution is the exact finite-resource static and adaptive optimization for the $[[5, 1, 3]]$ code inside the declared transversal ancilla-free measurement grammar; the four-class static collapse and normalizer characterization; the algebraic $q = 2/3$ basis transition; the global Bellman certificate for one adaptive policy over the full interval; and the failure-record gauge equivalence. General static/adaptive loss thresholds remain prior art [24], as do ideal arbitrary-stabilizer logical-BSM bounds [27]; the repetition-code experiment of Kumar *et al.* is experimental prior art and motivation rather than the finite-erasure five-qubit theorem proved here [28].

What is established here.

1. **Static exhaustive theorem:** all $6^5 = 7776$ allowed static analyzer schemes collapse to exactly four rational success polynomials.
2. **Exact phase theorem:** only two classes reach the upper envelope and $P_H - P_L = \frac{7}{32}q^4(3q - 2)$, giving the unique nontrivial switch $q_c = 2/3$.
3. **Code-structural theorem:** the 576 high-survival schemes are exactly the 18 full-support nontrivial logical-Pauli normalizer words times the 32 resolved-half choices.
4. **Adaptive theorem:** one feedforward policy is globally optimal for all $q \in [0, 1]$, with $P_{\text{ad}}^*(q) = \frac{5}{4}q^3 + \frac{7}{4}q^4 - \frac{65}{32}q^5$; 313 200 Bellman comparisons reduce to twelve nonnegative gap polynomials.
5. **Control/gauge theorem:** feedforward eliminates rather than shifts the static $q = 2/3$ phase boundary, and the complete sign-resolved failure record is a stabilizer-gauge degree of freedom within the declared grammar.

2 Five-qubit Bell-parity model

2.1 Code and logical Bell labels

We use the standard $[[5, 1, 3]]$ stabilizer generators

$$\begin{aligned} g_1 &= XZZXI, & g_2 &= IXZZX, \\ g_3 &= XIXZZ, & g_4 &= ZXIXZ, \end{aligned} \quad (1)$$

with logical representatives

$$\bar{X} = XXXXX, \quad \bar{Z} = ZZZZZ. \quad (2)$$

For code blocks A and B , the logical Bell label is the ordered pair of eigenvalues of

$$L_X = \bar{X}_A \bar{X}_B, \quad L_Z = \bar{Z}_A \bar{Z}_B. \quad (3)$$

For physical pair i , let $x_i, z_i \in \mathbb{F}_2$ encode the eigenvalues of $X_i^A X_i^B$ and $Z_i^A Z_i^B$ through $(-1)^{x_i}$ and $(-1)^{z_i}$. Since

$Y \otimes Y = -(X \otimes X)(Z \otimes Z)$, define $y_i = 1 + x_i + z_i \pmod{2}$. The doubled code stabilizers impose

$$\begin{aligned} x_1 + z_2 + z_3 + x_4 &= 0, \\ x_2 + z_3 + z_4 + x_5 &= 0, \\ x_1 + x_3 + z_4 + z_5 &= 0, \\ z_1 + x_2 + x_4 + z_5 &= 0. \end{aligned} \quad (4)$$

Hence the compatible physical Bell-parity space $\Omega_0 \subset \mathbb{F}_2^{10}$ contains $2^{10-4} = 64$ points. The logical bits are

$$\ell_X = \sum_{i=1}^5 x_i, \quad \ell_Z = \sum_{i=1}^5 z_i \pmod{2}, \quad (5)$$

and each logical Bell label occurs on 16 points.

2.2 Physical Bell analyzer and erasure model

A physical analyzer on pair i is specified by

$$a_i = (P_i, r_i), \quad P_i \in \{X, Y, Z\}, \quad r_i \in \mathbb{F}_2, \quad (6)$$

where $P_i^A P_i^B$ is guaranteed and r_i identifies the parity half that the standard ancilla-free analyzer fully resolves after local Clifford/Pauli-frame rotation [7, 20, 27]. For an intact pair, the guaranteed parity is always learned; in the designated half the complementary Bell parity is also learned; otherwise the Bell state remains unresolved but the two local P_i outcomes are physically available.

Let every photon survive independently with probability η . A transversal physical BSM requires one photon from each block, so the intact-pair probability is

$$q = \eta^2. \quad (7)$$

If either photon is absent, the pair is a heralded erasure carrying no trusted Bell information. The five pair erasures are independent. Dark counts, finite indistinguishability, correlated loss, and source errors are outside the theorem.

3 Static exact classification

For a static scheme $a = (a_1, \dots, a_5)$ and a fixed surviving-pair set, every point of Ω_0 produces a classical record. A record succeeds exactly when all compatible points have the same (ℓ_X, ℓ_Z) . Let $s_k(a)$ be the success probability conditioned on exactly k intact physical pairs, averaged uniformly over the $\binom{5}{k}$ surviving subsets and compatible parity assignments. Then

$$P_a(q) = \sum_{k=0}^5 \binom{5}{k} s_k(a) q^k (1-q)^{5-k}. \quad (8)$$

All coefficients are rational.

Theorem 1 (Exact four-class collapse). *Within the declared 6^5 static grammar, all 7776 schemes produce exactly four distinct sector vectors and four success polynomials. In every class,*

$$s_0 = s_1 = s_2 = 0, \quad s_3 = \frac{1}{8}. \quad (9)$$

The remaining data are given in Table 1.

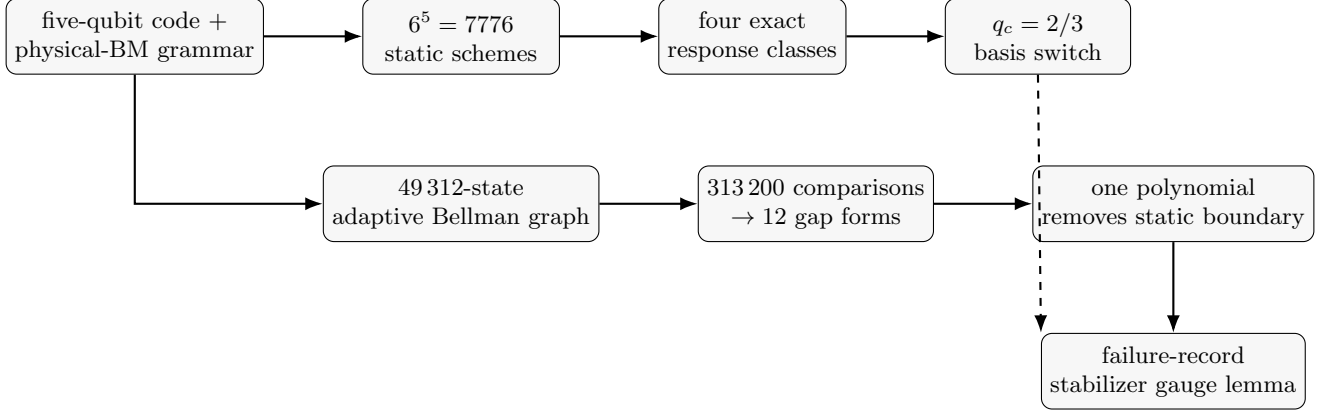


Figure 1: Proof architecture. The static calculation identifies an exact finite-resource phase boundary; the adaptive Bellman calculation proves that this boundary is induced by the static-control restriction; the gauge lemma then restores the complete physical failure record without changing the optimum.

Table 1: Complete exact static classification. Polynomial coefficients are in the ordinary power basis.

class	schemes	s_4	s_5	$P(q)$
$\mathcal{C}_H$	576	9/16	31/32	$\frac{5}{4}q^3 + \frac{5}{16}q^4 - \frac{19}{32}q^5$
$\mathcal{C}_G$	5760	47/80	7/8	$\frac{5}{4}q^3 + \frac{5}{16}q^4 - \frac{13}{16}q^5$
$\mathcal{C}_C$	720	23/40	13/16	$\frac{5}{4}q^3 + \frac{3}{8}q^4 - \frac{13}{16}q^5$
$\mathcal{C}_L$	720	13/20	3/4	$\frac{5}{4}q^3 + \frac{3}{4}q^4 - \frac{5}{4}q^5$

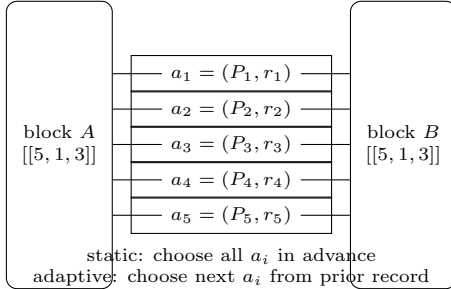


Figure 2: Common measurement grammar. Static protocols freeze all five analyzer choices; adaptive protocols may choose the next pair, basis, and resolved half after every heralded erasure or measurement outcome.

Proof sketch. Solver A enumerates the 64 solutions of Eq. (4), all 32 erasure subsets, and all 7776 static analyzer schemes using exact integer/rational arithmetic. Each measurement record partitions Ω_0 ; success means that both logical functionals in Eq. (5) are constant on a partition cell. Solver B independently represents ten-qubit Pauli operators as binary symplectic vectors and checks whether L_X, L_Z lie in the span of the code stabilizers together with the measured pair parities. The two implementations return the same four classes and multiplicities. The universal $s_3 = 1/8$ has a direct interpretation: with exactly three intact pairs, success requires full Bell resolution on all three survivors; no branch with at most two survivors spans both logical Bell operators.

3.1 The exact $q = 2/3$ basis transition

The two intermediate classes cannot reach the upper envelope. Exact subtraction gives

$$\begin{aligned}
 P_C(q) - P_G(q) &= -\frac{1}{16}q^4 < 0, \\
 P_H(q) - P_G(q) &= \frac{1}{32}q^4(7q - 4), \\
 P_G(q) - P_L(q) &= \frac{1}{16}q^4(7q - 5).
 \end{aligned} \tag{10}$$

Hence only $\mathcal{C}_H$ and $\mathcal{C}_L$ can be optimal.

Theorem 2 (Static basis-transition theorem). *The exact static optimum is*

$$P_{\text{static}}^*(q) = \begin{cases} \frac{5}{4}q^3 + \frac{3}{4}q^4 - \frac{5}{4}q^5, & 0 \leq q \leq \frac{2}{3}, \\ \frac{5}{4}q^3 + \frac{5}{16}q^4 - \frac{19}{32}q^5, & \frac{2}{3} \leq q \leq 1. \end{cases} \tag{11}$$

The two branches satisfy

$$P_H(q) - P_L(q) = \frac{7}{32}q^4(3q - 2), \tag{12}$$

so the unique nontrivial transition is

$$q_c = \frac{2}{3}, \quad \eta_c = \sqrt{\frac{2}{3}} = 0.816496580927\dots \tag{13}$$

At the transition $P_{\text{static}}^*(2/3) = 86/243$.

Proof sketch. After the intermediate classes are eliminated, the upper envelope is the maximum of two explicit rational polynomials. Their difference is Eq. (12); on $0 < q \leq 1$ its sign is the sign of $3q - 2$. No numerical root isolation is required.

3.2 Mechanism of the static transition

The two envelope classes agree at $k = 3$ but trade one-erasure robustness against intact efficiency:

$$(s_3, s_4, s_5)_H = \left(\frac{1}{8}, \frac{9}{16}, \frac{31}{32}\right), \quad (14)$$

$$(s_3, s_4, s_5)_L = \left(\frac{1}{8}, \frac{13}{20}, \frac{3}{4}\right). \quad (15)$$

Therefore

$$\begin{aligned} P_H - P_L &= 5q^4(1 - q) \left(\frac{9}{16} - \frac{13}{20}\right) + q^5 \left(\frac{31}{32} - \frac{3}{4}\right) \\ &= \frac{7}{32}q^4(3q - 2). \end{aligned} \quad (16)$$

Thus $q = 2/3$ is the exact balance point between improving the one-erasure sector and preserving the best fully intact performance.

4 Normalizer structure of the high-survival class

The four-class collapse has a code-theoretic explanation. For a static basis pattern define

$$W = P_1 P_2 P_3 P_4 P_5 \in \{X, Y, Z\}^5. \quad (17)$$

Let S be the five-qubit stabilizer group and $N(S)$ its Pauli normalizer.

Theorem 3 (Normalizer characterization). *A static basis word belongs to $\mathcal{C}_H$ if and only if W is a full-support representative of a nontrivial logical Pauli,*

$$W \in N(S) \setminus S. \quad (18)$$

There are exactly 18 such full-support words. For each word all $2^5 = 32$ choices of resolved Bell halves remain in $\mathcal{C}_H$, giving $18 \times 32 = 576$ schemes.

Proof sketch. For the fully intact branch, the five guaranteed pair parities multiply to a doubled Pauli word $W_A W_B$. If $W \in N(S) \setminus S$, this product is a nontrivial logical Pauli-pair parity. Whenever any one physical analyzer also resolves its complementary parity, the doubled stabilizers propagate sufficient information to determine the second logical Bell label. The only intact failure pattern is therefore the branch in which all five analyzers fail to reveal their complementary parity, giving $1 - 2^{-5} = 31/32$. Direct normalizer enumeration finds exactly 18 full-support nontrivial logical representatives. Conversely, basis words outside this set do not supply a logical parity through the guaranteed products and fall into the other three response classes. The full list is given in Appendix A.

5 Exact adaptive Bellman problem

5.1 Information states

A coarse-record information state is

$$\mathbf{s} = (U, \Omega), \quad (19)$$

where $U \subseteq \{1, \dots, 5\}$ is the set of unmeasured pairs and $\Omega \subseteq \Omega_0$ is the affine set of Bell assignments consistent with the record. A state is logically solved when both functionals in Eq. (5) are constant on Ω .

An independent representation stores only a canonical affine system

$$Mv = b, \quad v = (x_1, z_1, \dots, x_5, z_5)^T \in \mathbb{F}_2^{10}, \quad (20)$$

with the augmented matrix in reduced row-echelon form. Logical identifiability is equivalent to the two logical rows lying in the row span of the homogeneous code constraints together with the measured rows.

5.2 Bellman recurrence

For action $a = (i, P, r)$, let $V(\mathbf{s}; q)$ be the optimal eventual success probability. Then

$$Q_a(\mathbf{s}; q) = (1 - q)V(\mathbf{s}_E; q) + q \sum_{o \in \mathcal{O}_a} p(o | \mathbf{s}, a)V(\mathbf{s}_o; q), \quad (21)$$

and

$$V(\mathbf{s}; q) = \max_a Q_a(\mathbf{s}; q). \quad (22)$$

Every fixed-policy value is a rational polynomial in q of degree at most five.

Proposition 1 (Finite exact state graph). *The direct assignment solver and the independent affine-RREF solver each generate exactly 49 312 information states, of which 36 941 are nonterminal. At every state they agree on logical identifiability, exact branch probabilities, and the optimal value polynomial.*

Proof sketch. Solver A canonicalizes (U, Ω) directly. Solver B canonicalizes only the RREF of the augmented affine system. Each transition removes one physical pair, so the graph is finite and acyclic. The two state representations are generated independently; equality is checked locally at every state, not inferred merely from equal total counts.

6 Adaptive optimum and global certificate

Theorem 4 (Adaptive five-qubit success law). *Within the declared adaptive grammar,*

$$P_{\text{ad}}^*(q) = \frac{5}{4}q^3 + \frac{7}{4}q^4 - \frac{65}{32}q^5, \quad (23)$$

for every $0 \leq q \leq 1$. Equivalently,

$$(s_0, s_1, s_2, s_3, s_4, s_5) = \left(0, 0, 0, \frac{1}{8}, \frac{17}{20}, \frac{31}{32}\right). \quad (24)$$

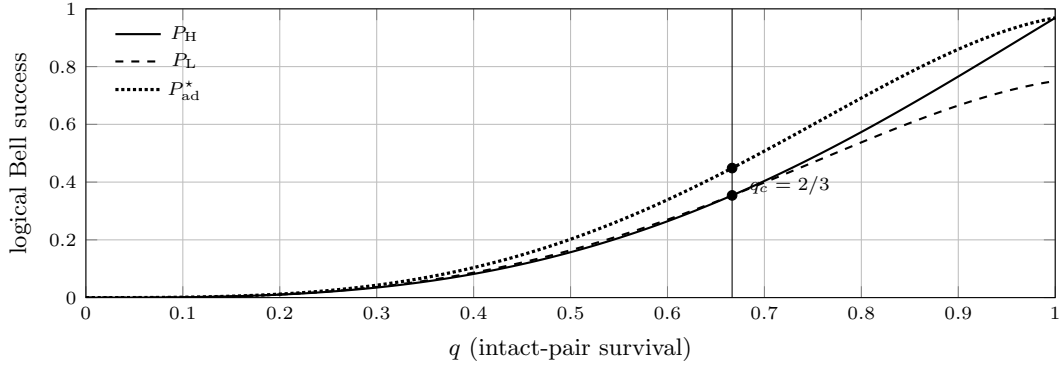


Figure 3: Static-to-adaptive phase structure. The two static envelope branches cross exactly at $q = 2/3$. The adaptive optimum is a single polynomial lying strictly above the static envelope throughout $0 < q < 1$ and meeting the lossless high-survival branch only at $q = 1$.

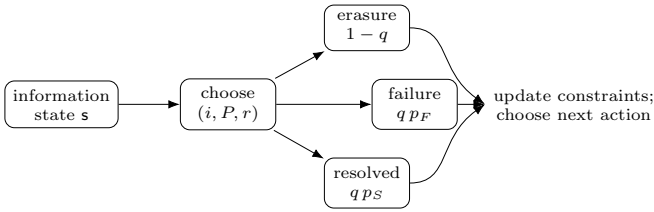


Figure 4: Adaptive decision grammar. The next physical pair, guaranteed Pauli basis, and resolved half may depend on the complete earlier record. Exact rational branch probabilities make the Bellman value a polynomial in q .

Proof sketch. The exact Bellman solve returns the Bernstein decomposition

$$P_{\text{ad}}^*(q) = 10 \left(\frac{1}{8} \right) q^3(1-q)^2 + 5 \left(\frac{17}{20} \right) q^4(1-q) + \frac{31}{32} q^5. \quad (25)$$

Expanding gives Eq. (23). The exact affine-span audit proves that $k \leq 2$ cannot determine both logical bits. At $k = 3$, successful branches require complete information on all three surviving pairs, yielding $s_3 = 1/8$. The adaptive gain is concentrated in the one-erasure sector, where feedforward raises success to $17/20$ while retaining the lossless $31/32$ endpoint known from the ideal five-qubit construction [27].

6.1 One policy is optimal for all q

Freeze one candidate policy π^* and compare it with every alternative action at every nonterminal state:

$$D_{s,a}(q) = Q_{\pi^*(s)}(s; q) - Q_a(s; q). \quad (26)$$

Theorem 5 (Global Bellman certificate). *Across all 313200 action comparisons, $D_{s,a}(q)$ assumes only the twelve forms in Table 2. Every form is nonnegative on $[0, 1]$. Therefore the same policy π^* is globally optimal for every physical survival probability.*

Proof sketch. Proceed backward in the number of remaining pairs. Terminal states are trivial. Assuming descendant values are optimal, nonnegativity of every local gap forces the frozen action to attain the Bellman

Table 2: Complete Bellman-gap alphabet.

$D(q)$	count	$D(q)$	count
0	241200	$q/2$	11520
q	2880	$q^2/4$	14400
$q^2/2$	14400	$q(1-q)/2$	11520
$q(1-q)$	5760	$q^3/8$	1440
$q^3/4$	720	$3q^3/8$	720
$q^2(1-q)/4$	2880	$3q^2(1-q)/4$	5760

maximum at the current state. Every nonzero gap is a positive rational multiple of a power of q , optionally times $(1-q)$; hence all are nonnegative on $[0, 1]$ without root isolation or floating-point interval arithmetic. Backward induction proves global optimality. The 241200 zero gaps encode policy degeneracy, not uncertainty in the root optimum. A representative optimal policy has only 220 reachable decision nodes and 640 reachable states including terminal branches.

7 Feedforward removes the static phase boundary

Theorem 6 (Static-to-adaptive phase theorem). *For every $0 < q < 1$,*

$$P_{\text{ad}}^*(q) > P_{\text{static}}^*(q). \quad (27)$$

More precisely,

$$P_{\text{ad}}^*(q) - P_{\text{H}}(q) = \frac{23}{16} q^4(1-q), \quad (28)$$

$$P_{\text{ad}}^*(q) - P_{\text{L}}(q) = q^4 \left(1 - \frac{25}{32} q \right). \quad (29)$$

Thus adaptivity removes, rather than shifts, the static $q = 2/3$ basis transition.

Proof. Subtract the two static envelope branches in Eq. (11) from Eq. (23). Equation (28) is positive on $0 < q < 1$; Eq. (29) is positive because $1 - 25q/32 \geq 7/32$ on $[0, 1]$. $\square$

At the former static transition,

$$P_{\text{static}}^*\left(\frac{2}{3}\right) = \frac{86}{243}, \quad P_{\text{ad}}^*\left(\frac{2}{3}\right) = \frac{109}{243}, \quad (30)$$

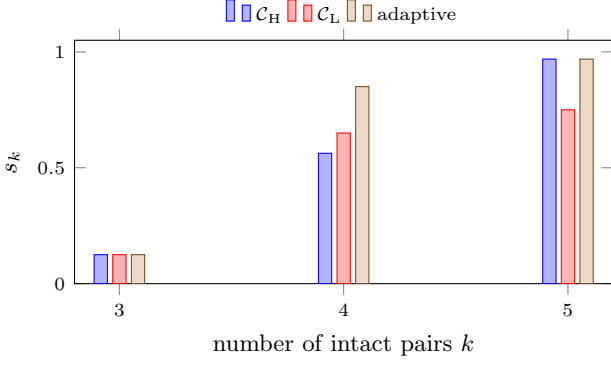


Figure 5: Sector mechanism. Adaptivity leaves the three-survivor and optimal intact sectors unchanged while lifting the one-erasure sector to $17/20$. This removes the static tradeoff responsible for the $q = 2/3$ crossing.

so

$$\Delta P = \frac{23}{243} = 0.0946502 \dots, \quad (31)$$

a relative increase of about 26.7% over the static optimum.

7.1 Where the adaptive gain comes from

The sector vectors expose the mechanism:

$$(s_3, s_4, s_5)_H = \left(\frac{1}{8}, \frac{9}{16}, \frac{31}{32} \right), \quad (32)$$

$$(s_3, s_4, s_5)_L = \left(\frac{1}{8}, \frac{13}{20}, \frac{3}{4} \right), \quad (33)$$

$$(s_3, s_4, s_5)_{ad} = \left(\frac{1}{8}, \frac{17}{20}, \frac{31}{32} \right). \quad (34)$$

Static measurement choice is forced to trade one-erasure robustness against intact efficiency. Feedforward conditions later bases on earlier information and simultaneously realizes the best intact endpoint and the strongest one-erasure sector.

8 Full optical failure record and gauge equivalence

The coarse Bellman model records only the guaranteed pair parity on an unresolved physical BM. A real analyzer also retains the two individual local P outcomes. We now show that the extra sign bit is operationally redundant in the declared adaptive grammar.

Lemma 1 (Failure-record stabilizer gauge). *For every physical pair i and guaranteed basis $P \in \{X, Y, Z\}$, there exists $g \in \langle g_1, g_2, g_3, g_4 \rangle$ whose local factor g_i anticommutes with P . Let*

$$G = g_A \otimes g_B. \quad (35)$$

Then G stabilizes every encoded logical Bell state, maps the two sign-resolved failure records at fixed product as

$$(a, b) \longleftrightarrow (-a, -b), \quad (36)$$

and commutes with every future doubled pair-parity observable $Q_j^A Q_j^B$ on unmeasured pair j .

Table 3: One stabilizer witness g with $\{g_i, P\} = 0$ for each pair and guaranteed basis.

pair i	$P = X$	$P = Y$	$P = Z$
1	g_4	g_1	g_1
2	g_1	g_2	g_2
3	g_1	g_3	g_3
4	g_2	g_1	g_1
5	g_3	g_2	g_2

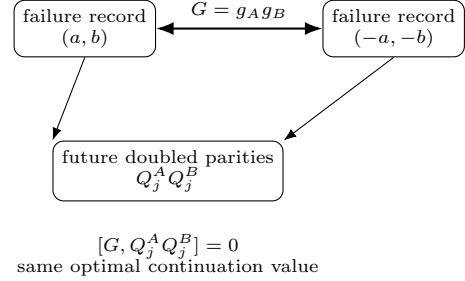


Figure 6: Failure-record gauge equivalence. The doubled code stabilizer flips both local failure signs while preserving the complete future pair-parity algebra, so the extra sign record cannot improve logical discrimination.

Proof sketch. Because g stabilizes each code block, $G = g_A g_B$ stabilizes the encoded Bell subspace. If $\{g_i, P\} = 0$, conjugation flips each local P outcome, so $(a, b) \mapsto (-a, -b)$ while preserving the product ab . On a future pair j , conjugation of Q_j by g_j may contribute a minus sign on one block, but the same sign appears on the other block and squares to $+1$. Thus G preserves the entire future pair-parity measurement algebra. Explicit low-weight witnesses are listed in Table 3.

Theorem 7 (Failure-record gauge equivalence). *Within the declared adaptive five-qubit grammar,*

$$P_{\text{full record}}^*(q) = P_{\text{coarse}}^*(q) = P_{\text{ad}}^*(q). \quad (37)$$

Thus retaining the complete local failure record cannot increase the optimal logical-Bell success probability.

Proof sketch. The gauge lemma pairs the two sign-resolved failure branches by a symmetry preserving every allowable future pair-parity measurement. Therefore any continuation policy after (a, b) can be conjugated into an equally successful continuation after $(-a, -b)$, so their Bellman continuation values are equal. Averaging the two branches loses no value, making the coarse parity record a value-preserving quotient. Equivalently, the two sign-resolved projectors at fixed product sum to

$$\Pi_s^{(P \otimes P)} = \frac{1}{2}(I + s P_A P_B), \quad (38)$$

which is exactly the coarse pair-parity projector; after tracing the measured pair, the unresolved local sign is the gauge degree of freedom identified above. Induction over subsequent failure nodes proves Eq. (37).

9 Independent verification and proof-carrying release

The static result was independently checked by two exact solvers: an affine-record partition over the 64 Bell-parity assignments and a binary symplectic span solver. They agree on all 7776 schemes, all four classes, and every survivor-sector probability. The complete static certificate contains 7776 rows.

The adaptive result was independently reconstructed in two different state representations. The direct solver propagates subsets of Ω_0 ; the second solver propagates canonical affine constraints in RREF over $\mathbb{F}_2$. They agree on the 49 312 states, 36 941 nonterminal states, the root optimum, all sector probabilities, and the multiset of Bellman-gap polynomials.

A separate full-instrument statevector replay retains the complete sign-resolved failure record. At five rational survival probabilities it reproduces the exact adaptive law to floating-point roundoff:

q	exact $P_{\text{ad}}^*(q)$	statevector replay
1/3	463/7776	0.05954218106995882
1/2	207/1024	0.20214843749999999
2/3	109/243	0.4485596707818928
3/4	19629/32768	0.5990295410156249
1	31/32	0.9687499999999998

The absolute discrepancies are 2.8×10^{-17} to 2.2×10^{-16} .

10 Relation to the recent experiment and established loss thresholds

The experimental result of Kumar, Reiss, van Loock, and Barz [28] and the present theorem address complementary but distinct levels of the logical-BSM problem. Their work demonstrates a logical BSM beyond the bare linear-optical limit using a two-qubit repetition/quantum-parity encoding. Here the object is instead the non-CSS distance-three five-qubit code, and the target is an exact finite-erasure optimization over static and adaptive analyzer architectures. In particular, the $7776 \rightarrow 4$ static collapse, the algebraic $q = 2/3$ basis switch, the globally optimal adaptive polynomial, and the failure-record gauge equivalence are not claims about their experimental encoding; they are the additional finite-code structure established in the present work.

The numerical appearance of $2/3$ requires care. Hilaire *et al.* derive a general static logical-LOBSM loss condition that yields $2/3$ for an unassisted physical BSM of success probability $1/2$ [24]. The present $q = 2/3$ is not that theorem: it is the crossing of two exact finite-code success polynomials in a fixed non-CSS measurement grammar. The equality of the two numbers is not used in any proof.

Likewise, the adaptive result is not a finite-code red-erivation of a general asymptotic threshold. $P_{\text{ad}}^*(q)$ is a per-attempt success law for one fixed five-qubit block and is nonzero for any $q > 0$. The lossless endpoint $P_{\text{ad}}^*(1) = 31/32$ agrees with the ideal arbitrary-stabilizer five-qubit result [27], while the finite-erasure polynomial and the removal of the static basis transition are the results established here.

11 Discussion

Four structural conclusions emerge. First, the static finite-erasure problem is dramatically compressible: 7776 physical analyzer schemes collapse to four exact response classes. Second, the high-survival class is not an arbitrary numerical cluster; it is exactly the full-support nontrivial logical-Pauli part of the five-qubit normalizer. Third, the static $q = 2/3$ switch is a resource-induced phase boundary: it balances one-erasure robustness against intact efficiency, and it vanishes when measurement bases may respond to the accumulated record. Fourth, physically available classical detector information need not be operationally useful at the logical level; in this code and grammar the extra local failure sign is a stabilizer-gauge coordinate.

These observations suggest concrete extensions. One may ask whether other non-CSS codes admit a small static response alphabet, whether the Bellman-gap alphabet remains finite and simple for Steane, surface, tree, or quantum-parity codes, and whether the gauge criterion admits a general code-theoretic characterization. Unequal pair survivals q_i would replace Eq. (8) by a multilinear reliability polynomial and may turn the single static crossing into a switching hypersurface. Dark counts, mode mismatch, imperfect interference, detector inefficiency beyond heralded erasure, source multiphoton components, and correlated loss remain outside the present theorem.

12 Conclusion

For the non-CSS five-qubit $[[5, 1, 3]]$ code under transversal ancilla-free $X/Y/Z$ -guaranteed linear-optical Bell measurements and independent pair erasure, the complete static-to-adaptive phase structure is exact. Static exhaustion yields four rational success classes and an algebraic basis transition at $q = 2/3$. Adaptive feedforward has the single global optimum

$$P_{\text{ad}}^*(q) = \frac{5}{4}q^3 + \frac{7}{4}q^4 - \frac{65}{32}q^5,$$

proved by two independent dynamic programs and a Bellman audit in which 313 200 action comparisons reduce to twelve manifestly nonnegative polynomial gaps. Feedforward preserves the lossless $31/32$ endpoint while raising the one-erasure sector to $17/20$, thereby eliminating the static phase boundary throughout $0 < q < 1$. A doubled-stabilizer gauge lemma then proves that the complete sign-resolved failure record does not improve the optimum. Relative to the recent repetition-code experiment of Kumar, Reiss, van Loock, and Barz [28], the contribution brought here is therefore not another demonstration of encoded Bell measurement, but an exact static-to-adaptive finite-erasure theory for a genuine distance-three non-CSS code, including its phase structure, global control optimum, and gauge-redundancy theorem. The resulting theorem is finite, proof-carrying, code-specific, and complementary to the established general theory of optical Bell limits and loss thresholds.

A Full-support logical basis words

The 18 full-support words defining the high-survival class are

$$\begin{array}{lll} XXXXX & XXZYZ & XYYXZ \\ XYZZY & XZXYY & XZYZX \\ YXYZZ & YXZZY & YYXZZ \\ YYYYY & YZXXZ & YZZYX \\ ZXXZY & ZXYXX & ZYXYZ \\ ZYZXX & ZZYXY & ZZZZZ \end{array}$$

Each commutes with the five-qubit stabilizer group, is not itself a stabilizer, and has no identity factor. All 32 choices of resolved parity halves for each word remain in $\mathcal{C}_H$.

B Certificate identities

At the static transition,

$$P_H\left(\frac{2}{3}\right) = P_L\left(\frac{2}{3}\right) = \frac{86}{243}, \quad (39)$$

while

$$P_G\left(\frac{2}{3}\right) = \frac{85}{243}, \quad P_C\left(\frac{2}{3}\right) = \frac{82}{243}. \quad (40)$$

The exact class counts satisfy

$$576 + 5760 + 720 + 720 = 7776. \quad (41)$$

The Bellman comparison multiplicities satisfy

$$\begin{aligned} 313200 &= 241200 + 11520 + 2880 + 14400 \\ &+ 14400 + 11520 + 5760 + 1440 \\ &+ 720 + 720 + 2880 + 5760. \end{aligned} \quad (42)$$

Every nonzero Bellman gap is generated by powers of q , the factor $(1 - q)$, and positive rational constants.

C Gauge-witness verification

The witness table follows from the local Pauli letters in the four generators. For example, at pair 1, g_4 has local factor Z and anticommutes with X , while g_1 has local factor X and anticommutes with both Y and Z . The remaining entries are checked identically. Since g_{AGB} contains the same local Pauli on both code blocks, it commutes with every doubled future Pauli $Q_j^A Q_j^B$ irrespective of whether g_j commutes or anticommutes with Q_j on one block.

Computational Verification Statement

All computational algorithms, proof-search procedures, and certificate criteria were developed by the author(s). Large Language Models (LLMs) such as ChatGPT and Claude were utilized strictly as computational assistants for debugging code and cross-checking routines. No mathematical claims rely on LLM outputs; all results are grounded exclusively in independently reproducible, rigorously verified certificates (exact, symbolic, rational, interval, exhaustive, or controlled high-precision).

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